

Filter Summary Report: CG,TIA,Automated,simple,Z3,Z4

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Contents

1 Examined $H(z)$ for CG TIA Automated simple Z3 Z4: $\frac{Z_3 Z_4 g_m}{2Z_3 g_m + Z_4 g_m}$

$$H(z) = \frac{Z_3 Z_4 g_m}{2Z_3 g_m + Z_4 g_m}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(\infty, \infty, R_3, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_4 R_3 s}{2C_4 L_4 R_3 s^2 + L_4 s + 2R_3}$$

Parameters:

Q: $\frac{2\sqrt{C_4} R_3}{\sqrt{L_4}}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{1}{2C_4 R_3}$
 K-LP: 0
 K-HP: 0
 K-BP: R_3
 Qz: None
 Wz: None

3.2 BP-2 $Z(s) = \left(\infty, \infty, R_3, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{L_4 R_3 R_4 s}{2C_4 L_4 R_3 R_4 s^2 + 2R_3 R_4 + s(2L_4 R_3 + L_4 R_4)}$$

Parameters:

Q: $\frac{2\sqrt{C_4} R_3 R_4}{2\sqrt{L_4} R_3 + \sqrt{L_4} R_4}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{2\sqrt{L_4} R_3 + \sqrt{L_4} R_4}{2C_4 \sqrt{L_4} R_3 R_4}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_3 R_4}{2R_3 + R_4}$
 Qz: None
 Wz: None

3.3 BP-3 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{L_4 R_4 s}{2L_4 s + 2R_4 + s^2(C_3 L_4 R_4 + 2C_4 L_4 R_4)}$$

Parameters:

Q: $\frac{\sqrt{2} R_4 \sqrt{C_3 + 2C_4}}{2\sqrt{L_4}}$
 wo: $\frac{\sqrt{2}}{\sqrt{C_3 L_4 + 2C_4 L_4}}$
 bandwidth: $\frac{2\sqrt{L_4}}{R_4 \sqrt{C_3 + 2C_4} \sqrt{C_3 L_4 + 2C_4 L_4}}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_4}{2}$
 Qz: None
 Wz: None

3.4 BP-4 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{2} R_3 \sqrt{C_3 + 2C_4}}{\sqrt{L_4}}$
 wo: $\frac{\sqrt{2}}{\sqrt{C_3 L_4 + 2C_4 L_4}}$
 bandwidth: $\frac{\sqrt{L_4}}{R_3 \sqrt{C_3 + 2C_4} \sqrt{C_3 L_4 + 2C_4 L_4}}$
 K-LP: 0
 K-HP: 0
 K-BP: R_3
 Qz: None
 Wz: None

$$H(s) = \frac{L_4 R_3 s}{L_4 s + 2R_3 + s^2 (C_3 L_4 R_3 + 2C_4 L_4 R_3)}$$

3.5 BP-5 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{2} R_3 R_4 \sqrt{C_3 + 2C_4}}{2\sqrt{L_4 R_3 + \sqrt{L_4} R_4}}$
 wo: $\frac{\sqrt{2}}{\sqrt{C_3 L_4 + 2C_4 L_4}}$
 bandwidth: $\frac{2\sqrt{L_4} R_3 + \sqrt{L_4} R_4}{R_3 R_4 \sqrt{C_3 + 2C_4} \sqrt{C_3 L_4 + 2C_4 L_4}}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_3 R_4}{2R_3 + R_4}$
 Qz: None
 Wz: None

$$H(s) = \frac{L_4 R_3 R_4 s}{2R_3 R_4 + s^2 (C_3 L_4 R_3 R_4 + 2C_4 L_4 R_3 R_4) + s (2L_4 R_3 + L_4 R_4)}$$

3.6 BP-6 $Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{C_3} R_4}{2\sqrt{L_3}}$
 wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
 bandwidth: $\frac{2}{C_3 R_4}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_4}{2}$
 Qz: None
 Wz: None

$$H(s) = \frac{L_3 R_4 s}{C_3 L_3 R_4 s^2 + 2L_3 s + R_4}$$

3.7 BP-7 $Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

Parameters:

Q: $\frac{R_4 \sqrt{C_3 + 2C_4}}{2\sqrt{L_3}}$
 wo: $\frac{1}{\sqrt{C_3 L_3 + 2C_4 L_3}}$
 bandwidth: $\frac{2\sqrt{L_3}}{R_4 \sqrt{C_3 + 2C_4} \sqrt{C_3 L_3 + 2C_4 L_3}}$

$$H(s) = \frac{L_3 R_4 s}{2L_3 s + R_4 + s^2 (C_3 L_3 R_4 + 2C_4 L_3 R_4)}$$

K-LP: 0
K-HP: 0
K-BP: $\frac{R_4}{2}$
Qz: None
Wz: None

$$\mathbf{3.8 \quad BP-8} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \infty, \quad \infty \right)$$

Parameters:

Q: $\frac{R_4 \sqrt{C_3 + 2C_4} \sqrt{2L_3 + L_4}}{2\sqrt{L_3} \sqrt{L_4}}$
wo: $\frac{\sqrt{2L_3 + L_4}}{\sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4}}$
bandwidth: $\frac{2\sqrt{L_3} \sqrt{L_4}}{R_4 \sqrt{C_3 + 2C_4} \sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4}}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_4}{2}$
Qz: None
Wz: None

$$H(s) = \frac{L_3 L_4 R_4 s}{2L_3 L_4 s + 2L_3 R_4 + L_4 R_4 + s^2 (C_3 L_3 L_4 R_4 + 2C_4 L_3 L_4 R_4)}$$

$$\mathbf{3.9 \quad BP-9} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad R_4, \quad \infty, \quad \infty \right)$$

Parameters:

Q: $\frac{\sqrt{C_3} R_3 R_4}{2\sqrt{L_3} R_3 + \sqrt{L_3} R_4}$
wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
bandwidth: $\frac{2\sqrt{L_3} R_3 + \sqrt{L_3} R_4}{C_3 \sqrt{L_3} R_3 R_4}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_3 R_4}{2R_3 + R_4}$
Qz: None
Wz: None

$$H(s) = \frac{L_3 R_3 R_4 s}{C_3 L_3 R_3 R_4 s^2 + R_3 R_4 + s (2L_3 R_3 + L_3 R_4)}$$

$$\mathbf{3.10 \quad BP-10} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

Parameters:

Q: $\frac{R_3 \sqrt{C_3 + 2C_4}}{\sqrt{L_3}}$
wo: $\frac{1}{\sqrt{C_3 L_3 + 2C_4 L_3}}$
bandwidth: $\frac{\sqrt{L_3}}{R_3 \sqrt{C_3 + 2C_4} \sqrt{C_3 L_3 + 2C_4 L_3}}$
K-LP: 0
K-HP: 0
K-BP: R_3
Qz: None
Wz: None

$$H(s) = \frac{L_3 R_3 s}{L_3 s + R_3 + s^2 (C_3 L_3 R_3 + 2C_4 L_3 R_3)}$$

3.11 BP-11 $Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_3 R_3 R_4 s}{R_3 R_4 + s^2 (C_3 L_3 R_3 R_4 + 2C_4 L_3 R_3 R_4) + s (2L_3 R_3 + L_3 R_4)}$$

Parameters:

Q: $\frac{R_3 R_4 \sqrt{C_3 + 2C_4}}{2\sqrt{L_3} R_3 + \sqrt{L_3} R_4}$
 wo: $\frac{1}{\sqrt{C_3 L_3 + 2C_4 L_3}}$
 bandwidth: $\frac{2\sqrt{L_3} R_3 + \sqrt{L_3} R_4}{R_3 R_4 \sqrt{C_3 + 2C_4} \sqrt{C_3 L_3 + 2C_4 L_3}}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_3 R_4}{2R_3 + R_4}$
 Qz: None
 Wz: None

3.12 BP-12 $Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{L_3 L_4 R_3 s}{L_3 L_4 s + 2L_3 R_3 + L_4 R_3 + s^2 (C_3 L_3 L_4 R_3 + 2C_4 L_3 L_4 R_3)}$$

Parameters:

Q: $\frac{R_3 \sqrt{C_3 + 2C_4} \sqrt{2L_3 + L_4}}{\sqrt{L_3} \sqrt{L_4}}$
 wo: $\frac{\sqrt{2L_3 + L_4}}{\sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4}}$
 bandwidth: $\frac{\sqrt{L_3} \sqrt{L_4}}{R_3 \sqrt{C_3 + 2C_4} \sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4}}$
 K-LP: 0
 K-HP: 0
 K-BP: R_3
 Qz: None
 Wz: None

3.13 BP-13 $Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$

$$H(s) = \frac{L_3 L_4 R_3 R_4 s}{2L_3 R_3 R_4 + L_4 R_3 R_4 + s^2 (C_3 L_3 L_4 R_3 R_4 + 2C_4 L_3 L_4 R_3 R_4) + s (2L_3 L_4 R_3 + L_3 L_4 R_4)}$$

Parameters:

Q: $\frac{R_3 R_4 \sqrt{C_3 + 2C_4} \sqrt{2L_3 + L_4}}{2\sqrt{L_3} \sqrt{L_4} R_3 + \sqrt{L_3} \sqrt{L_4} R_4}$
 wo: $\frac{\sqrt{2L_3 + L_4}}{\sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4}}$
 bandwidth: $\frac{2\sqrt{L_3} \sqrt{L_4} R_3 + \sqrt{L_3} \sqrt{L_4} R_4}{R_3 R_4 \sqrt{C_3 + 2C_4} \sqrt{C_3 L_3 L_4 + 2C_4 L_3 L_4}}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_3 R_4}{2R_3 + R_4}$
 Qz: None
 Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(\infty, \infty, R_3, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 R_3 s^2 + R_3}{C_4 L_4 s^2 + 2C_4 R_3 s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_4}}{2\sqrt{C_4}R_3}$
 wo: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$
 bandwidth: $\frac{2R_3}{L_4}$
 K-LP: R_3
 K-HP: R_3
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$

$$4.2 \quad \text{BS-2} \quad Z(s) = \left(\infty, \infty, R_3, \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4L_4R_3R_4s^2 + R_3R_4}{2C_4R_3R_4s + 2R_3 + R_4 + s^2(2C_4L_4R_3 + C_4L_4R_4)}$$

Parameters:

Q: $\frac{2\sqrt{L_4}R_3+\sqrt{L_4}R_4}{2\sqrt{C_4}R_3R_4}$
 wo: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$
 bandwidth: $\frac{2R_3R_4}{\sqrt{L_4}(2\sqrt{L_4}R_3+\sqrt{L_4}R_4)}$
 K-LP: $\frac{R_3R_4}{2R_3+R_4}$
 K-HP: $\frac{R_3R_4}{2R_3+R_4}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$

$$4.3 \quad \text{BS-3} \quad Z(s) = \left(\infty, \infty, L_3s + \frac{1}{C_3s}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{C_3L_3R_4s^2 + R_4}{2C_3L_3s^2 + C_3R_4s + 2}$$

Parameters:

Q: $\frac{2\sqrt{L_3}}{\sqrt{C_3}R_4}$
 wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
 bandwidth: $\frac{R_4}{2L_3}$
 K-LP: $\frac{R_4}{2}$
 K-HP: $\frac{R_4}{2}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

$$4.4 \quad \text{BS-4} \quad Z(s) = \left(\infty, \infty, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{C_3L_3R_3R_4s^2 + R_3R_4}{C_3R_3R_4s + 2R_3 + R_4 + s^2(2C_3L_3R_3 + C_3L_3R_4)}$$

Parameters:

Q: $\frac{2\sqrt{L_3}R_3+\sqrt{L_3}R_4}{\sqrt{C_3}R_3R_4}$
 wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
 bandwidth: $\frac{R_3R_4}{\sqrt{L_3}(2\sqrt{L_3}R_3+\sqrt{L_3}R_4)}$
 K-LP: $\frac{R_3R_4}{2R_3+R_4}$
 K-HP: $\frac{R_3R_4}{2R_3+R_4}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

5 GE

5.1 GE-1 $Z(s) = \left(\infty, \infty, R_3, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 R_3 s^2 + C_4 R_3 R_4 s + R_3}{C_4 L_4 s^2 + s(2C_4 R_3 + C_4 R_4) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{L_4}}{2\sqrt{C_4 R_3} + \sqrt{C_4 R_4}} \\ \text{wo: } & \frac{1}{\sqrt{C_4} \sqrt{L_4}} \\ \text{bandwidth: } & \frac{2\sqrt{C_4 R_3} + \sqrt{C_4 R_4}}{\sqrt{C_4} L_4} \\ \text{K-LP: } & R_3 \\ \text{K-HP: } & R_3 \\ \text{K-BP: } & \frac{R_3 R_4}{2R_3 + R_4} \\ \text{Qz: } & \frac{\sqrt{L_4}}{\sqrt{C_4} R_4} \\ \text{Wz: } & \frac{1}{\sqrt{C_4} \sqrt{L_4}} \end{aligned}$$

5.2 GE-2 $Z(s) = \left(\infty, \infty, R_3, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_4 L_4 R_3 R_4 s^2 + L_4 R_3 s + R_3 R_4}{L_4 s + 2R_3 + R_4 + s^2(2C_4 L_4 R_3 + C_4 L_4 R_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_4 R_3} + \sqrt{C_4 R_4}}{\sqrt{L_4}} \\ \text{wo: } & \frac{1}{\sqrt{C_4} \sqrt{L_4}} \\ \text{bandwidth: } & \frac{1}{\sqrt{C_4}(2\sqrt{C_4 R_3} + \sqrt{C_4 R_4})} \\ \text{K-LP: } & \frac{R_3 R_4}{2R_3 + R_4} \\ \text{K-HP: } & \frac{R_3 R_4}{2R_3 + R_4} \\ \text{K-BP: } & R_3 \\ \text{Qz: } & \frac{\sqrt{C_4} R_4}{\sqrt{L_4}} \\ \text{Wz: } & \frac{1}{\sqrt{C_4} \sqrt{L_4}} \end{aligned}$$

5.3 GE-3 $Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_3 R_4 s^2 + C_3 R_3 R_4 s + R_4}{2C_3 L_3 s^2 + s(2C_3 R_3 + C_3 R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{L_3}}{2\sqrt{C_3 R_3} + \sqrt{C_3 R_4}} \\ \text{wo: } & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \\ \text{bandwidth: } & \frac{2\sqrt{C_3 R_3} + \sqrt{C_3 R_4}}{2\sqrt{C_3} L_3} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & \frac{R_4}{2} \\ \text{K-BP: } & \frac{R_3 R_4}{2R_3 + R_4} \\ \text{Qz: } & \frac{\sqrt{L_3}}{\sqrt{C_3} R_3} \\ \text{Wz: } & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \end{aligned}$$

5.4 GE-4 $Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_3 L_3 R_3 R_4 s^2 + L_3 R_4 s + R_3 R_4}{2 L_3 s + 2 R_3 + R_4 + s^2 (2 C_3 L_3 R_3 + C_3 L_3 R_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_3}R_3 + \sqrt{C_3}R_4}{2\sqrt{L_3}} \\ \text{wo: } & \frac{1}{\sqrt{C_3}\sqrt{L_3}} \\ \text{bandwidth: } & \frac{2}{\sqrt{C_3}(2\sqrt{C_3}R_3 + \sqrt{C_3}R_4)} \\ \text{K-LP: } & \frac{R_3 R_4}{2R_3 + R_4} \\ \text{K-HP: } & \frac{R_3 R_4}{2R_3 + R_4} \\ \text{K-BP: } & \frac{R_4}{2} \\ \text{Qz: } & \frac{\sqrt{C_3}R_3}{\sqrt{L_3}} \\ \text{Wz: } & \frac{1}{\sqrt{C_3}\sqrt{L_3}} \end{aligned}$$

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_4 R_3 R_4 s + R_3}{C_3 C_4 R_3 R_4 s^2 + s (C_3 R_3 + 2 C_4 R_3 + C_4 R_4) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}{C_3 R_3 + 2 C_4 R_3 + C_4 R_4} \\ \text{wo: } & \frac{1}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}} \\ \text{bandwidth: } & \frac{C_3 R_3 + 2 C_4 R_3 + C_4 R_4}{C_3 C_4 R_3 R_4} \\ \text{K-LP: } & R_3 \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_4 R_3 R_4}{C_3 R_3 + 2 C_4 R_3 + C_4 R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_3 R_3 R_4 s + R_4}{2 C_3 C_4 R_3 R_4 s^2 + s (2 C_3 R_3 + C_3 R_4 + 2 C_4 R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}{2 C_3 R_3 + C_3 R_4 + 2 C_4 R_4} \\ \text{wo: } & \frac{1}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}} \\ \text{bandwidth: } & \frac{2 C_3 R_3 + C_3 R_4 + 2 C_4 R_4}{2 C_3 C_4 R_3 R_4} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_3 R_3 R_4}{2 C_3 R_3 + C_3 R_4 + 2 C_4 R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty, \infty, R_3, R_4, \infty, \infty)$

$$H(s) = \frac{R_3 R_4}{2R_3 + R_4}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty, \infty, R_3, \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{R_3}{2C_4 R_3 s + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty, \infty, R_3, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty\right)$

$$H(s) = \frac{R_3 R_4}{2C_4 R_3 R_4 s + 2R_3 + R_4}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty, \infty, R_3, R_4 + \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{C_4 R_3 R_4 s + R_3}{s(2C_4 R_3 + C_4 R_4) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, R_4, \infty, \infty\right)$

$$H(s) = \frac{R_4}{C_3 R_4 s + 2}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{1}{s(C_3 + 2C_4)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty\right)$

$$H(s) = \frac{R_4}{s(C_3 R_4 + 2C_4 R_4) + 2}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{C_4 R_4 s + 1}{C_3 C_4 R_4 s^2 + s(C_3 + 2C_4)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{C_4 L_4 s^2 + 1}{C_3 C_4 L_4 s^3 + s(C_3 + 2C_4)}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_4 s}{s^2 (C_3 L_4 + 2C_4 L_4) + 2}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{1}{C_3 s}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 s^2 + C_4 R_4 s + 1}{C_3 C_4 L_4 s^3 + C_3 C_4 R_4 s^2 + s (C_3 + 2C_4)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_3 C_4 L_4 R_4 s^3 + C_3 R_4 s + s^2 (C_3 L_4 + 2C_4 L_4) + 2}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{1}{C_3 s}, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_4 s^2 + R_4}{C_3 C_4 L_4 R_4 s^3 + 2C_4 L_4 s^2 + s (C_3 R_4 + 2C_4 R_4) + 2}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4}{C_3 R_3 R_4 s + 2R_3 + R_4}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_3}{s (C_3 R_3 + 2C_4 R_3) + 1}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_3 R_4}{2R_3 + R_4 + s (C_3 R_3 R_4 + 2C_4 R_3 R_4)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_3 s^2 + R_3}{C_3 C_4 L_4 R_3 s^3 + C_4 L_4 s^2 + s (C_3 R_3 + 2C_4 R_3) + 1}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_3 s^2 + C_4 R_3 R_4 s + R_3}{C_3 C_4 L_4 R_3 s^3 + s^2 (C_3 C_4 R_3 R_4 + C_4 L_4) + s (C_3 R_3 + 2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\infty, \quad \infty, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_3 R_4 s^2 + L_4 R_3 s + R_3 R_4}{C_3 C_4 L_4 R_3 R_4 s^3 + 2R_3 + R_4 + s^2 (C_3 L_4 R_3 + 2C_4 L_4 R_3 + C_4 L_4 R_4) + s (C_3 R_3 R_4 + L_4)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_3 R_4 s^2 + R_3 R_4}{C_3 C_4 L_4 R_3 R_4 s^3 + 2R_3 + R_4 + s^2 (2C_4 L_4 R_3 + C_4 L_4 R_4) + s (C_3 R_3 R_4 + 2C_4 R_3 R_4)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4, \infty, \infty \right)$$

$$H(s) = \frac{C_3 R_3 R_4 s + R_4}{s (2C_3 R_3 + C_3 R_4) + 2}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 R_3 s + 1}{2C_3 C_4 R_3 s^2 + s (C_3 + 2C_4)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 R_3 R_4 s^2 + s (C_3 R_3 + C_4 R_4) + 1}{s^2 (2C_3 C_4 R_3 + C_3 C_4 R_4) + s (C_3 + 2C_4)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_3 s^3 + C_3 R_3 s + C_4 L_4 s^2 + 1}{C_3 C_4 L_4 s^3 + 2C_3 C_4 R_3 s^2 + s (C_3 + 2C_4)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_4 R_3 s^2 + L_4 s}{2C_3 C_4 L_4 R_3 s^3 + 2C_3 R_3 s + s^2 (C_3 L_4 + 2C_4 L_4) + 2}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_3 s^3 + s^2 (C_3 C_4 R_3 R_4 + C_4 L_4) + s (C_3 R_3 + C_4 R_4) + 1}{C_3 C_4 L_4 s^3 + s^2 (2C_3 C_4 R_3 + C_3 C_4 R_4) + s (C_3 + 2C_4)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_4 R_3 R_4 s^2 + L_4 R_4 s}{2C_3 C_4 L_4 R_3 R_4 s^3 + 2R_4 + s^2 (2C_3 L_4 R_3 + C_3 L_4 R_4 + 2C_4 L_4 R_4) + s (2C_3 R_3 R_4 + 2L_4)}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_3 R_4 s^3 + R_4 + s^2 (C_3 L_4 R_3 + C_4 L_4 R_4) + s (C_3 R_3 R_4 + L_4)}{s^3 (2C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4) + s^2 (C_3 L_4 + 2C_4 L_4) + s (2C_3 R_3 + C_3 R_4) + 2}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_4 R_3 R_4 s^3 + C_3 R_3 R_4 s + C_4 L_4 R_4 s^2 + R_4}{s^3 (2C_3 C_4 L_4 R_3 + C_3 C_4 L_4 R_4) + s^2 (2C_3 C_4 R_3 R_4 + 2C_4 L_4) + s (2C_3 R_3 + C_3 R_4 + 2C_4 R_4) + 2}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 s^2 + 1}{2C_3 C_4 L_3 s^3 + s (C_3 + 2C_4)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_4 s^2 + R_4}{2C_3 C_4 L_3 R_4 s^3 + 2C_3 L_3 s^2 + s (C_3 R_4 + 2C_4 R_4) + 2}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_4 s^3 + C_3 L_3 s^2 + C_4 R_4 s + 1}{2C_3 C_4 L_3 s^3 + C_3 C_4 R_4 s^2 + s (C_3 + 2C_4)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 s^4 + s^2 (C_3 L_3 + C_4 L_4) + 1}{s^3 (2C_3 C_4 L_3 + C_3 C_4 L_4) + s (C_3 + 2C_4)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 s^3 + L_4 s}{2C_3 C_4 L_3 L_4 s^4 + s^2 (2C_3 L_3 + C_3 L_4 + 2C_4 L_4) + 2}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 s^4 + C_3 C_4 L_3 R_4 s^3 + C_4 R_4 s + s^2 (C_3 L_3 + C_4 L_4) + 1}{C_3 C_4 R_4 s^2 + s^3 (2C_3 C_4 L_3 + C_3 C_4 L_4) + s (C_3 + 2C_4)}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_4 s^3 + L_4 R_4 s}{2C_3 C_4 L_3 L_4 R_4 s^4 + 2C_3 L_3 L_4 s^3 + 2L_4 s + 2R_4 + s^2 (2C_3 L_3 R_4 + C_3 L_4 R_4 + 2C_4 L_4 R_4)}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_4 s^4 + C_3 L_3 L_4 s^3 + L_4 s + R_4 + s^2 (C_3 L_3 R_4 + C_4 L_4 R_4)}{2C_3 C_4 L_3 L_4 s^4 + C_3 C_4 L_4 R_4 s^3 + C_3 R_4 s + s^2 (2C_3 L_3 + C_3 L_4 + 2C_4 L_4) + 2}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_4 s^4 + R_4 + s^2 (C_3 L_3 R_4 + C_4 L_4 R_4)}{2 C_3 C_4 L_3 L_4 s^4 + s^3 (2 C_3 C_4 L_3 R_4 + C_3 C_4 L_4 R_4) + s^2 (2 C_3 L_3 + 2 C_4 L_4) + s (C_3 R_4 + 2 C_4 R_4) + 2}$$

$$\mathbf{9.39 \quad X-INVALID-ORDER-39} \quad Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{L_3 s}{s^2 (C_3 L_3 + 2 C_4 L_3) + 1}$$

$$\mathbf{9.40 \quad X-INVALID-ORDER-40} \quad Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 R_4 s^2 + L_3 s}{C_3 C_4 L_3 R_4 s^3 + C_4 R_4 s + s^2 (C_3 L_3 + 2 C_4 L_3) + 1}$$

$$\mathbf{9.41 \quad X-INVALID-ORDER-41} \quad Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 s^3 + L_3 s}{C_3 C_4 L_3 L_4 s^4 + s^2 (C_3 L_3 + 2 C_4 L_3 + C_4 L_4) + 1}$$

$$\mathbf{9.42 \quad X-INVALID-ORDER-42} \quad Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{L_3 L_4 s}{2 L_3 + L_4 + s^2 (C_3 L_3 L_4 + 2 C_4 L_3 L_4)}$$

$$\mathbf{9.43 \quad X-INVALID-ORDER-43} \quad Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 s^3 + C_4 L_3 R_4 s^2 + L_3 s}{C_3 C_4 L_3 L_4 s^4 + C_3 C_4 L_3 R_4 s^3 + C_4 R_4 s + s^2 (C_3 L_3 + 2 C_4 L_3 + C_4 L_4) + 1}$$

$$\mathbf{9.44 \quad X-INVALID-ORDER-44} \quad Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_4 s^3 + L_3 L_4 s^2 + L_3 R_4 s}{C_3 C_4 L_3 L_4 R_4 s^4 + R_4 + s^3 (C_3 L_3 L_4 + 2 C_4 L_3 L_4) + s^2 (C_3 L_3 R_4 + C_4 L_4 R_4) + s (2 L_3 + L_4)}$$

$$\mathbf{9.45 \quad X-INVALID-ORDER-45} \quad Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_4 s^3 + L_3 R_4 s}{C_3 C_4 L_3 L_4 R_4 s^4 + 2 C_4 L_3 L_4 s^3 + 2 L_3 s + R_4 + s^2 (C_3 L_3 R_4 + 2 C_4 L_3 R_4 + C_4 L_4 R_4)}$$

$$\mathbf{9.46 \quad X-INVALID-ORDER-46} \quad Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 s^2 + C_3 R_3 s + 1}{2 C_3 C_4 L_3 s^3 + 2 C_3 C_4 R_3 s^2 + s (C_3 + 2 C_4)}$$

9.47 X-INVALID-ORDER-47 $Z(s) = \left(\infty, \infty, L_3s + R_3 + \frac{1}{C_3s}, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{C_3L_3R_4s^2 + C_3R_3R_4s + R_4}{2C_3C_4L_3R_4s^3 + s^2(2C_3C_4R_3R_4 + 2C_3L_3) + s(2C_3R_3 + C_3R_4 + 2C_4R_4) + 2}$$

9.48 X-INVALID-ORDER-48 $Z(s) = \left(\infty, \infty, L_3s + R_3 + \frac{1}{C_3s}, R_4 + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_3C_4L_3R_4s^3 + s^2(C_3C_4R_3R_4 + C_3L_3) + s(C_3R_3 + C_4R_4) + 1}{2C_3C_4L_3s^3 + s^2(2C_3C_4R_3 + C_3C_4R_4) + s(C_3 + 2C_4)}$$

9.49 X-INVALID-ORDER-49 $Z(s) = \left(\infty, \infty, L_3s + R_3 + \frac{1}{C_3s}, L_4s + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_3C_4L_3L_4s^4 + C_3C_4L_4R_3s^3 + C_3R_3s + s^2(C_3L_3 + C_4L_4) + 1}{2C_3C_4R_3s^2 + s^3(2C_3C_4L_3 + C_3C_4L_4) + s(C_3 + 2C_4)}$$

9.50 X-INVALID-ORDER-50 $Z(s) = \left(\infty, \infty, L_3s + R_3 + \frac{1}{C_3s}, \frac{L_4s}{C_4L_4s^2+1}, \infty, \infty \right)$

$$H(s) = \frac{C_3L_3L_4s^3 + C_3L_4R_3s^2 + L_4s}{2C_3C_4L_3L_4s^4 + 2C_3C_4L_4R_3s^3 + 2C_3R_3s + s^2(2C_3L_3 + C_3L_4 + 2C_4L_4) + 2}$$

9.51 X-INVALID-ORDER-51 $Z(s) = \left(\infty, \infty, L_3s + R_3 + \frac{1}{C_3s}, L_4s + R_4 + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_3C_4L_3L_4s^4 + s^3(C_3C_4L_3R_4 + C_3C_4L_4R_3) + s^2(C_3C_4R_3R_4 + C_3L_3 + C_4L_4) + s(C_3R_3 + C_4R_4) + 1}{s^3(2C_3C_4L_3 + C_3C_4L_4) + s^2(2C_3C_4R_3 + C_3C_4R_4) + s(C_3 + 2C_4)}$$

9.52 X-INVALID-ORDER-52 $Z(s) = \left(\infty, \infty, L_3s + R_3 + \frac{1}{C_3s}, \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_3L_3L_4R_4s^3 + C_3L_4R_3R_4s^2 + L_4R_4s}{2C_3C_4L_3L_4R_4s^4 + 2R_4 + s^3(2C_3C_4L_4R_3R_4 + 2C_3L_3L_4) + s^2(2C_3L_3R_4 + 2C_3L_4R_3 + C_3L_4R_4 + 2C_4L_4R_4) + s(2C_3R_3R_4 + 2L_4)}$$

9.53 X-INVALID-ORDER-53 $Z(s) = \left(\infty, \infty, L_3s + R_3 + \frac{1}{C_3s}, \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \infty, \infty \right)$

$$H(s) = \frac{C_3C_4L_3L_4R_4s^4 + R_4 + s^3(C_3C_4L_4R_3R_4 + C_3L_3L_4) + s^2(C_3L_3R_4 + C_3L_4R_3 + C_4L_4R_4) + s(C_3R_3R_4 + L_4)}{2C_3C_4L_3L_4s^4 + s^3(2C_3C_4L_4R_3 + C_3C_4L_4R_4) + s^2(2C_3L_3 + C_3L_4 + 2C_4L_4) + s(2C_3R_3 + C_3R_4) + 2}$$

9.54 X-INVALID-ORDER-54 $Z(s) = \left(\infty, \infty, L_3s + R_3 + \frac{1}{C_3s}, \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{C_3C_4L_3L_4R_4s^4 + C_3C_4L_4R_3R_4s^3 + C_3R_3R_4s + R_4 + s^2(C_3L_3R_4 + C_4L_4R_4)}{2C_3C_4L_3L_4s^4 + s^3(2C_3C_4L_3R_4 + 2C_3C_4L_4R_3 + C_3C_4L_4R_4) + s^2(2C_3C_4R_3R_4 + 2C_3L_3 + 2C_4L_4) + s(2C_3R_3 + C_3R_4 + 2C_4R_4) + 2}$$

9.55 X-INVALID-ORDER-55 $Z(s) = \left(\infty, \infty, \frac{L_3R_3s}{C_3L_3R_3s^2+L_3s+R_3}, R_4 + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_4L_3R_3R_4s^2 + L_3R_3s}{C_3C_4L_3R_3R_4s^3 + R_3 + s^2(C_3L_3R_3 + 2C_4L_3R_3 + C_4L_3R_4) + s(C_4R_3R_4 + L_3)}$$

$$\mathbf{9.56 \quad X-INVALID-ORDER-56} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_3 s^3 + L_3 R_3 s}{C_3 C_4 L_3 L_4 R_3 s^4 + C_4 L_3 L_4 s^3 + L_3 s + R_3 + s^2 (C_3 L_3 R_3 + 2C_4 L_3 R_3 + C_4 L_4 R_3)}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_3 s^3 + C_4 L_3 R_3 R_4 s^2 + L_3 R_3 s}{C_3 C_4 L_3 L_4 R_3 s^4 + R_3 + s^3 (C_3 C_4 L_3 R_3 R_4 + C_4 L_3 L_4) + s^2 (C_3 L_3 R_3 + 2C_4 L_3 R_3 + C_4 L_3 R_4 + C_4 L_4 R_3) + s (C_4 R_3 R_4 + L_3)}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_3 R_4 s^3 + L_3 L_4 R_3 s^2 + L_3 R_3 R_4 s}{C_3 C_4 L_3 L_4 R_3 R_4 s^4 + R_3 R_4 + s^3 (C_3 L_3 L_4 R_3 + 2C_4 L_3 L_4 R_3 + C_4 L_3 L_4 R_4) + s^2 (C_3 L_3 R_3 R_4 + C_4 L_4 R_3 R_4 + L_3 L_4) + s (2L_3 R_3 + L_3 R_4 + L_4 R_3)}$$

$$\mathbf{9.59 \quad X-INVALID-ORDER-59} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_4 L_3 L_4 R_3 R_4 s^3 + L_3 R_3 R_4 s}{C_3 C_4 L_3 L_4 R_3 R_4 s^4 + R_3 R_4 + s^3 (2C_4 L_3 L_4 R_3 + C_4 L_3 L_4 R_4) + s^2 (C_3 L_3 R_3 R_4 + 2C_4 L_3 R_3 R_4 + C_4 L_4 R_3 R_4) + s (2L_3 R_3 + L_3 R_4)}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{2C_3 C_4 L_3 R_3 s^3 + 2C_4 R_3 s + s^2 (C_3 L_3 + 2C_4 L_3) + 1}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 R_4 s^2 + L_3 R_4 s + R_3 R_4}{2C_3 C_4 L_3 R_3 R_4 s^3 + 2R_3 + R_4 + s^2 (2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_3 R_4) + s (2C_4 R_3 R_4 + 2L_3)}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_3 R_4 s^3 + R_3 + s^2 (C_3 L_3 R_3 + C_4 L_3 R_4) + s (C_4 R_3 R_4 + L_3)}{s^3 (2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4) + s^2 (C_3 L_3 + 2C_4 L_3) + s (2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 s^4 + C_4 L_3 L_4 s^3 + L_3 s + R_3 + s^2 (C_3 L_3 R_3 + C_4 L_4 R_3)}{C_3 C_4 L_3 L_4 s^4 + 2C_3 C_4 L_3 R_3 s^3 + 2C_4 R_3 s + s^2 (C_3 L_3 + 2C_4 L_3 + C_4 L_4) + 1}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_3 s^3 + L_3 L_4 s^2 + L_4 R_3 s}{2C_3 C_4 L_3 L_4 R_3 s^4 + 2R_3 + s^3 (C_3 L_3 L_4 + 2C_4 L_3 L_4) + s^2 (2C_3 L_3 R_3 + 2C_4 L_4 R_3) + s (2L_3 + L_4)}$$

$$\mathbf{9.65 \quad X-INVALID-ORDER-65} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, L_4 s + R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 s^4 + R_3 + s^3 (C_3 C_4 L_3 R_3 R_4 + C_4 L_3 L_4) + s^2 (C_3 L_3 R_3 + C_4 L_3 R_4 + C_4 L_4 R_3) + s (C_4 R_3 R_4 + L_3)}{C_3 C_4 L_3 L_4 s^4 + s^3 (2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4) + s^2 (C_3 L_3 + 2C_4 L_3 + C_4 L_4) + s (2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.66 \quad X-INVALID-ORDER-66} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_3 R_4 s^3 + L_3 L_4 R_4 s^2 + L_4 R_3 R_4 s}{2C_3 C_4 L_3 L_4 R_3 R_4 s^4 + 2R_3 R_4 + s^3 (2C_3 L_3 L_4 R_3 + C_3 L_3 L_4 R_4 + 2C_4 L_3 L_4 R_4) + s^2 (2C_3 L_3 R_3 R_4 + 2C_4 L_4 R_3 R_4 + 2L_3 L_4) + s (2L_3 R_4 + 2L_4 R_3 + L_4 R_4)}$$

$$\mathbf{9.67 \quad X-INVALID-ORDER-67} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 R_4 s^4 + R_3 R_4 + s^3 (C_3 L_3 L_4 R_3 + C_4 L_3 L_4 R_4) + s^2 (C_3 L_3 R_3 R_4 + C_4 L_4 R_3 R_4 + L_3 L_4) + s (L_3 R_4 + L_4 R_3)}{2R_3 + R_4 + s^4 (2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4) + s^3 (C_3 L_3 L_4 + 2C_4 L_3 L_4) + s^2 (2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_4 R_3 + C_4 L_4 R_4) + s (2L_3 + L_4)}$$

$$\mathbf{9.68 \quad X-INVALID-ORDER-68} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 R_4 s^4 + C_4 L_3 L_4 R_4 s^3 + L_3 R_4 s + R_3 R_4 + s^2 (C_3 L_3 R_3 R_4 + C_4 L_4 R_3 R_4)}{2R_3 + R_4 + s^4 (2C_3 C_4 L_3 L_4 R_3 + C_3 C_4 L_3 L_4 R_4) + s^3 (2C_3 C_4 L_3 R_3 R_4 + 2C_4 L_3 L_4) + s^2 (2C_3 L_3 R_3 + C_3 L_3 R_4 + 2C_4 L_4 R_3 + C_4 L_4 R_4) + s (2C_4 R_3 R_4 + 2L_3)}$$

$$\mathbf{9.69 \quad X-INVALID-ORDER-69} \quad Z(s) = \left(\infty, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + R_3}{2C_3 C_4 L_3 R_3 s^3 + C_3 L_3 s^2 + s (C_3 R_3 + 2C_4 R_3) + 1}$$

$$\mathbf{9.70 \quad X-INVALID-ORDER-70} \quad Z(s) = \left(\infty, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 R_4 s^2 + R_3 R_4}{2C_3 C_4 L_3 R_3 R_4 s^3 + 2R_3 + R_4 + s^2 (2C_3 L_3 R_3 + C_3 L_3 R_4) + s (C_3 R_3 R_4 + 2C_4 R_3 R_4)}$$

$$\mathbf{9.71 \quad X-INVALID-ORDER-71} \quad Z(s) = \left(\infty, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 R_3 R_4 s^3 + C_3 L_3 R_3 s^2 + C_4 R_3 R_4 s + R_3}{s^3 (2C_3 C_4 L_3 R_3 + C_3 C_4 L_3 R_4) + s^2 (C_3 C_4 R_3 R_4 + C_3 L_3) + s (C_3 R_3 + 2C_4 R_3 + C_4 R_4) + 1}$$

$$\mathbf{9.72 \quad X-INVALID-ORDER-72} \quad Z(s) = \left(\infty, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, L_4 s + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 C_4 L_3 L_4 R_3 s^4 + R_3 + s^2 (C_3 L_3 R_3 + C_4 L_4 R_3)}{C_3 C_4 L_3 L_4 s^4 + s^3 (2C_3 C_4 L_3 R_3 + C_3 C_4 L_4 R_3) + s^2 (C_3 L_3 + C_4 L_4) + s (C_3 R_3 + 2C_4 R_3) + 1}$$

$$\mathbf{9.73 \quad X-INVALID-ORDER-73} \quad Z(s) = \left(\infty, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 L_4 R_3 s^3 + L_4 R_3 s}{2C_3 C_4 L_3 L_4 R_3 s^4 + C_3 L_3 L_4 s^3 + L_4 s + 2R_3 + s^2 (2C_3 L_3 R_3 + C_3 L_4 R_3 + 2C_4 L_4 R_3)}$$

9.74 X-INVALID-ORDER-74 $Z(s) = \left(\infty, \infty, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, L_4s + R_4 + \frac{1}{C_4s}, \infty, \infty \right)$

$$H(s) = \frac{C_3C_4L_3L_4R_3s^4 + C_3C_4L_3R_3R_4s^3 + C_4R_3R_4s + R_3 + s^2(C_3L_3R_3 + C_4L_4R_3)}{C_3C_4L_3L_4s^4 + s^3(2C_3C_4L_3R_3 + C_3C_4L_3R_4 + C_3C_4L_4R_3) + s^2(C_3C_4R_3R_4 + C_3L_3 + C_4L_4) + s(C_3R_3 + 2C_4R_3 + C_4R_4) + 1}$$

9.75 X-INVALID-ORDER-75 $Z(s) = \left(\infty, \infty, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \infty, \infty \right)$

$$H(s) = \frac{C_3L_3L_4R_3R_4s^3 + L_4R_3R_4s}{2C_3C_4L_3L_4R_3R_4s^4 + 2R_3R_4 + s^3(2C_3L_3L_4R_3 + C_3L_3L_4R_4) + s^2(2C_3L_3R_3R_4 + C_3L_4R_3R_4 + 2C_4L_4R_3R_4) + s(2L_4R_3 + L_4R_4)}$$

9.76 X-INVALID-ORDER-76 $Z(s) = \left(\infty, \infty, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1}, \infty, \infty \right)$

$$H(s) = \frac{C_3C_4L_3L_4R_3R_4s^4 + C_3L_3L_4R_3s^3 + L_4R_3s + R_3R_4 + s^2(C_3L_3R_3R_4 + C_4L_4R_3R_4)}{2R_3 + R_4 + s^4(2C_3C_4L_3L_4R_3 + C_3C_4L_3L_4R_4) + s^3(C_3C_4L_4R_3R_4 + C_3L_3L_4) + s^2(2C_3L_3R_3 + C_3L_3R_4 + C_3L_4R_3 + 2C_4L_4R_3 + C_4L_4R_4) + s(C_3R_3R_4 + L_4)}$$

9.77 X-INVALID-ORDER-77 $Z(s) = \left(\infty, \infty, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{C_3C_4L_3L_4R_3R_4s^4 + R_3R_4 + s^2(C_3L_3R_3R_4 + C_4L_4R_3R_4)}{2R_3 + R_4 + s^4(2C_3C_4L_3L_4R_3 + C_3C_4L_3L_4R_4) + s^3(2C_3C_4L_3R_3R_4 + C_3C_4L_4R_3R_4) + s^2(2C_3L_3R_3 + C_3L_3R_4 + 2C_4L_4R_3 + C_4L_4R_4) + s(C_3R_3R_4 + 2C_4R_3R_4)}$$

10 X-INVALID-WZ

11 X-PolynomialError